

Metabolomics homework

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LC–MS metabolomikai pipeline (v2, ST000816)

Lipid peak mátrix: neg + pos ionmód, minták × jellemzők. Lépések a NMR_Metabolomika_Pipeline_v2.ipynb szerkezetéhez igazítva:

QC → PQN/log1p/Pareto (+ klasszikus LC–MS előfeldolgozás) → multipanel PCA → univariáns + vulkán → heatmap → PLS-DA + VIP + S-plot → RF. A függvények a `src/metabolomics_*.py` modulokban vannak.

```
In [ ]: # Init gdrive and python environment
```

```
from google.colab import drive
drive.mount('/content/drive')

!git clone https://github.com/csambrus/MetabolKD.git
%cd /content/MetabolKD
!pip install -r ./requirements.txt
```

```
In [ ]: import sys
import traceback
from pathlib import Path
from datetime import datetime

PROJECT_ROOT = Path("/content/MetabolKD")

if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))

print("PROJECT_ROOT:", PROJECT_ROOT)
from src.runtime_setup import setup_tensorflow_runtime, setup_notebook_error_logger, set_global_seed

set_global_seed()
setup_notebook_error_logger()
setup_tensorflow_runtime()
```

1. Adatbetöltés (ST000816, baseline — mint a házi feladatban)

```
In [26]: from src.download_dataset import download_dataset, combine_pos_neg_to_feature_matrix

data = download_dataset()
X, meta = combine_pos_neg_to_feature_matrix(data)

print("NEG raw:", data["neg"].shape)
print("POS raw:", data["pos"].shape)
print("Feature matrix:", X.shape)
print(meta.head())

# Sample_id (index), progressor_status, visit: egyedi kategóriák száma
cat_counts = {
    "sample_id (index)": meta.index.nunique(),
    "progressor_status": meta["progressor_status"].nunique(dropna=False),
    "visit": meta["visit"].nunique(dropna=False),
}
print("Egyedi kategóriák száma:")
for k, n in cat_counts.items():
    print(f" {k}: {n}")
for col in ("progressor_status", "visit"):
    print(f"\n{col} – érték szerinti darabszám:")
    print(meta[col].value_counts(dropna=False))

# Visit: hiányzó és szöveges "NA" → Baseline; ezután csak Baseline (Followup kiesik)
meta["visit"] = meta["visit"].replace({"NA": "Baseline", "na": "Baseline"}).fillna("Baseline")

baseline_mask = meta["visit"] == "Baseline"
meta = meta.loc[baseline_mask]
X = X.loc[meta.index]

print(f"Baseline minták: {len(meta)} (X: {X.shape})")
print(meta["visit"].value_counts())

# LC-MS pipeline: a további cellák X_raw néven hivatkoznak a mátrixra
X_raw = X
```

```
GDRIVE_DATA: megvan /content/drive/MyDrive/MetabolKD/data/ST000816_AN001293_Results.txt
GDRIVE_DATA: megvan /content/drive/MyDrive/MetabolKD/data/ST000816_AN001294_Results.txt
Másolva → /content/MetabolKD/data/ST000816_AN001293_Results.txt
Másolva → /content/MetabolKD/data/ST000816_AN001294_Results.txt
NEG raw: (162, 101)
POS raw: (413, 101)
Feature matrix: (100, 573)
      progressor_status  visit
sample_id
S00017985      Non-progressor  Baseline
S00017986      Non-progressor  Followup
S00017987           Progressor  Baseline
S00017988           Progressor  Followup
S00017989      Non-progressor  Baseline
Egyedi kategóriák száma:
  sample_id (index): 100
  progressor_status: 2
  visit: 3
```

```
progressor_status – érték szerinti darabszám:
progressor_status
Non-progressor      50
Progressor          50
Name: count, dtype: int64
```

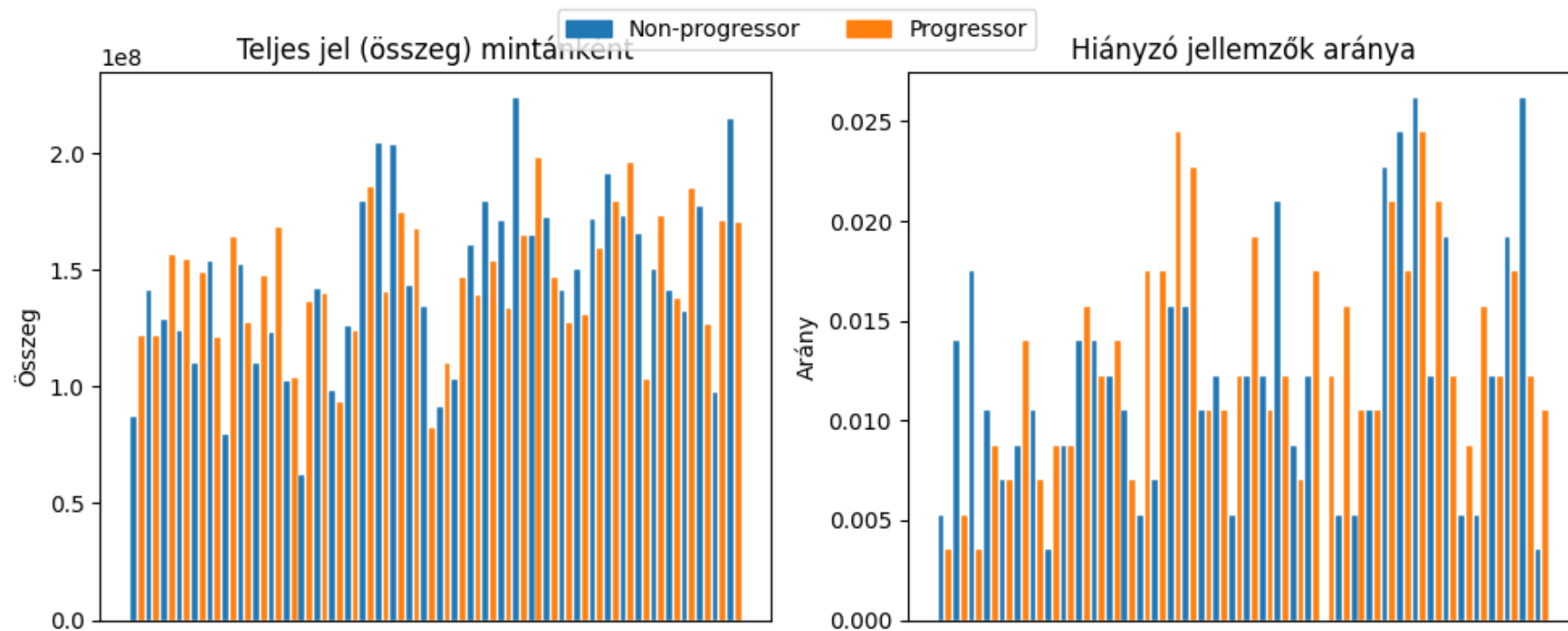
```
visit – érték szerinti darabszám:
visit
NA          60
Baseline    20
Followup    20
Name: count, dtype: int64
Baseline minták: 80 (X: (80, 573))
visit
Baseline     80
Name: count, dtype: int64
```

2. QC (mintánkénti összjel és hiányzás)

```
In [27]: from src.metabolomics_qc import qc_sample_table
         from src.metabolomics_plotting import plot_qc_bars

         qc = qc_sample_table(X_raw)
         print(qc.describe())
         fig = plot_qc_bars(qc, meta, hue_col="progressor_status")
         fig.savefig("qc_overview.png", dpi=150, bbox_inches="tight")
```

	total_signal	n_missing	missing_frac
count	8.000000e+01	80.000000	80.000000
mean	1.450142e+08	7.087500	0.012369
std	3.330350e+07	3.409569	0.005950
min	6.200004e+07	0.000000	0.000000
25%	1.238442e+08	5.000000	0.008726
50%	1.446944e+08	7.000000	0.012216
75%	1.703300e+08	9.000000	0.015707
max	2.236135e+08	15.000000	0.026178



3. NMR-stílusú mátrixok (PQN, log1p, Pareto) + klasszikus LC–MS előfeldolgozás

```
In [28]: from src.metabolomics_preprocessing import nmr_style_matrices, preprocess_feature_matrix
```

```
blocks = nmr_style_matrices(X_raw, impute_first=True)
for k, df in blocks.items():
    print(k, df.shape)
X_pareto = blocks["X_pareto"]
X_log = blocks["X_log"]

X_proc, steps = preprocess_feature_matrix(X_raw)
print("Klasszikus LC–MS lépések:", " → ".join(steps))
X_proc.head()
```

X_raw (80, 573)

X_pqn (80, 573)

X_log (80, 573)

X_pareto (80, 573)

Klasszikus LC–MS lépések: median_sample_norm → impute_half_min → log1p → pareto_columns

Out[28]:

	neg_CL 70:5; [M-2H] (2-)@6.12	neg_CL 74:1; [M-2H] (2-)@6.80	neg_CL 74:5; [M-2H] (2-)@6.59	neg_CL 74:7; [M-2H] (2-)@6.14	neg_CL 78:5; [M-2H] (2-)@6.48	neg_CL 82:13; [M-2H] (2-)@6.44	neg_CerP 34:1; [M- H]-@5.89	neg_IS PC 34:0; [M-Ac-H]-	neg_IS PE 34:0 ; [M-H]-	neg_IS PG 34:0 ; [M-H]-	...	pos_plasmenyl- PE 36:4; [M+H]+@6.72	pos_
sample_id													
S00017985	-0.056649	0.549517	-0.853816	-0.333168	0.567393	-0.047831	-0.032609	0.207037	0.226124	0.460968	...	-0.303224	
S00017987	-0.091050	-0.003848	0.319627	0.039181	0.609764	0.033419	0.006338	-0.581517	-0.463254	-0.550401	...	0.238637	
S00017989	-0.188057	0.335733	-0.239595	-0.131554	-0.348958	-0.314348	-0.231672	0.299410	0.375354	0.149090	...	0.781294	
S00017991	-0.506865	0.771575	-0.506659	-0.304283	0.079430	0.273572	0.263582	0.167796	0.121956	0.411802	...	0.258942	
S00017993	0.167618	0.576869	0.867470	0.511845	-0.393220	0.230193	0.285687	-0.089026	0.017502	0.314454	...	0.979717	

5 rows × 573 columns

4. PCA multipanel (Pareto + log1p térben, extra Z-score nélkül — mint az NMR X_PARETO)

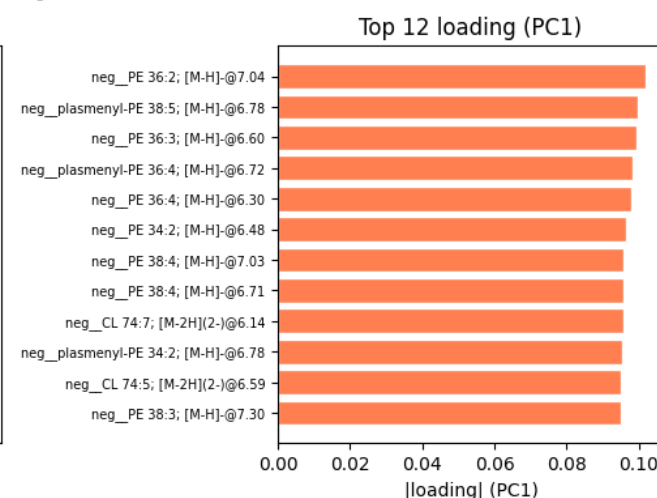
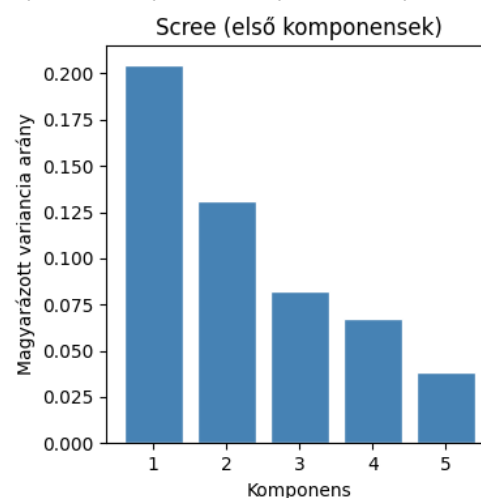
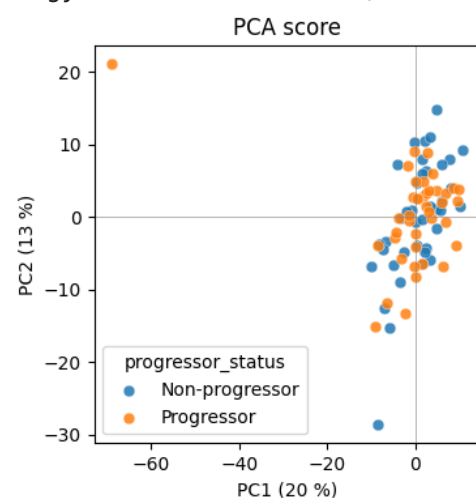
```
In [29]: from src.metabolomics_multivariate import fit_pca
from src.metabolomics_plotting import plot_pca_multipanel, plot_pca_scores

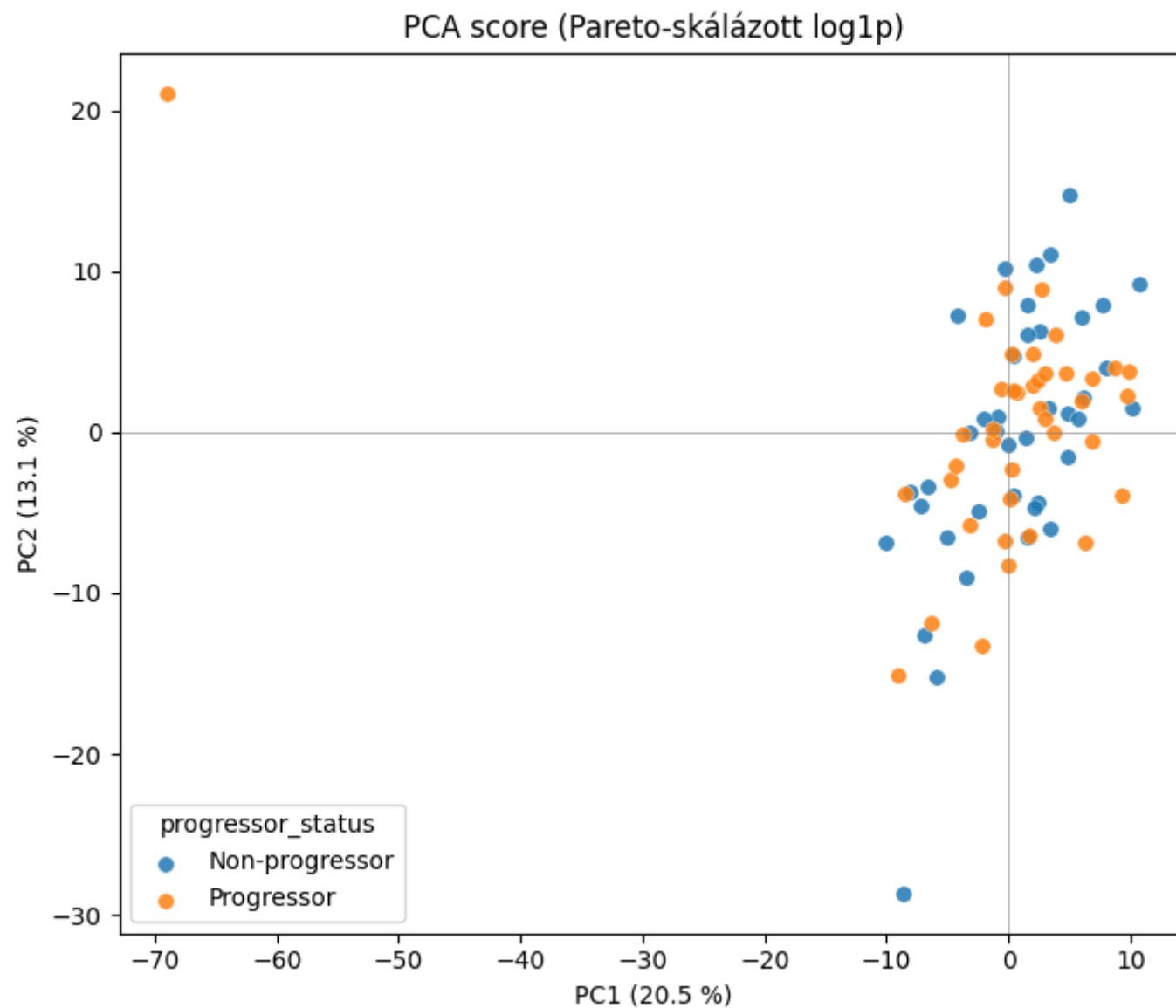
pca_out = fit_pca(X_pareto, n_components=5, standardize=False)
scores = pca_out["scores"]
var = pca_out["explained_variance_ratio"]
print("Magyarázott variancia (első 5 PC):", [round(float(v), 4) for v in var])

fig = plot_pca_multipanel(pca_out, meta, hue_col="progressor_status")
fig.savefig("pca_multipanel.png", dpi=150, bbox_inches="tight")

fig2 = plot_pca_scores(
    scores, meta,
    pc_x="PC1", pc_y="PC2",
    hue_col="progressor_status",
    explained=var,
    title="PCA score (Pareto-skálázott log1p)",
)
fig2.savefig("pca_scores.png", dpi=150, bbox_inches="tight")
```

Magyarázott variancia (első 5 PC): [0.2047, 0.1307, 0.0821, 0.0671, 0.0386]





5. Unvariáns elemzés + vulkán (Baseline: Progressor vs Non-progressor)

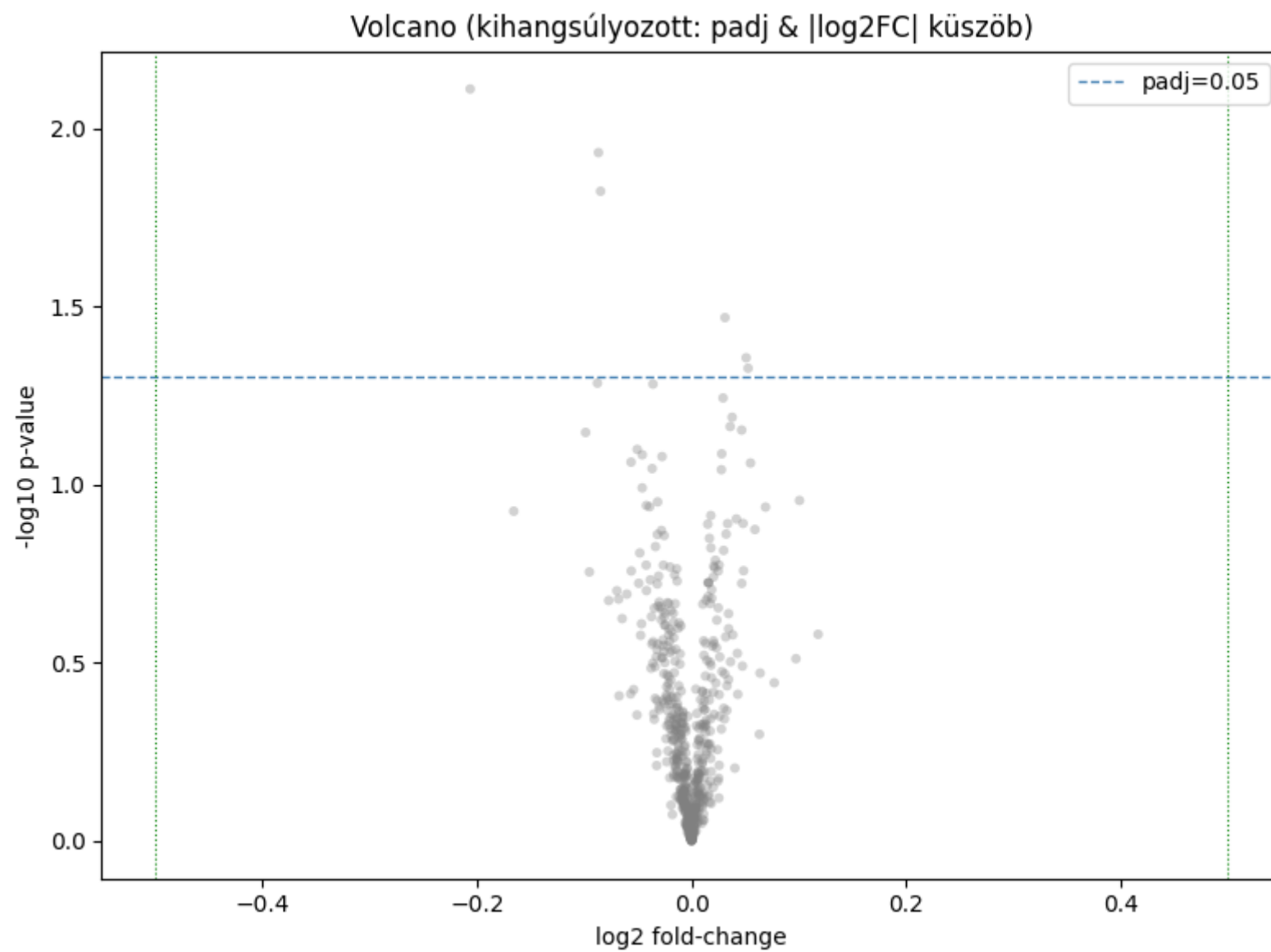
```
In [30]: # Csak baseline látogatás
bl = meta["visit"] == "Baseline"
Xb_log = X_log.loc[bl]
Xb_par = X_pareto.loc[bl]
metab = meta.loc[bl]

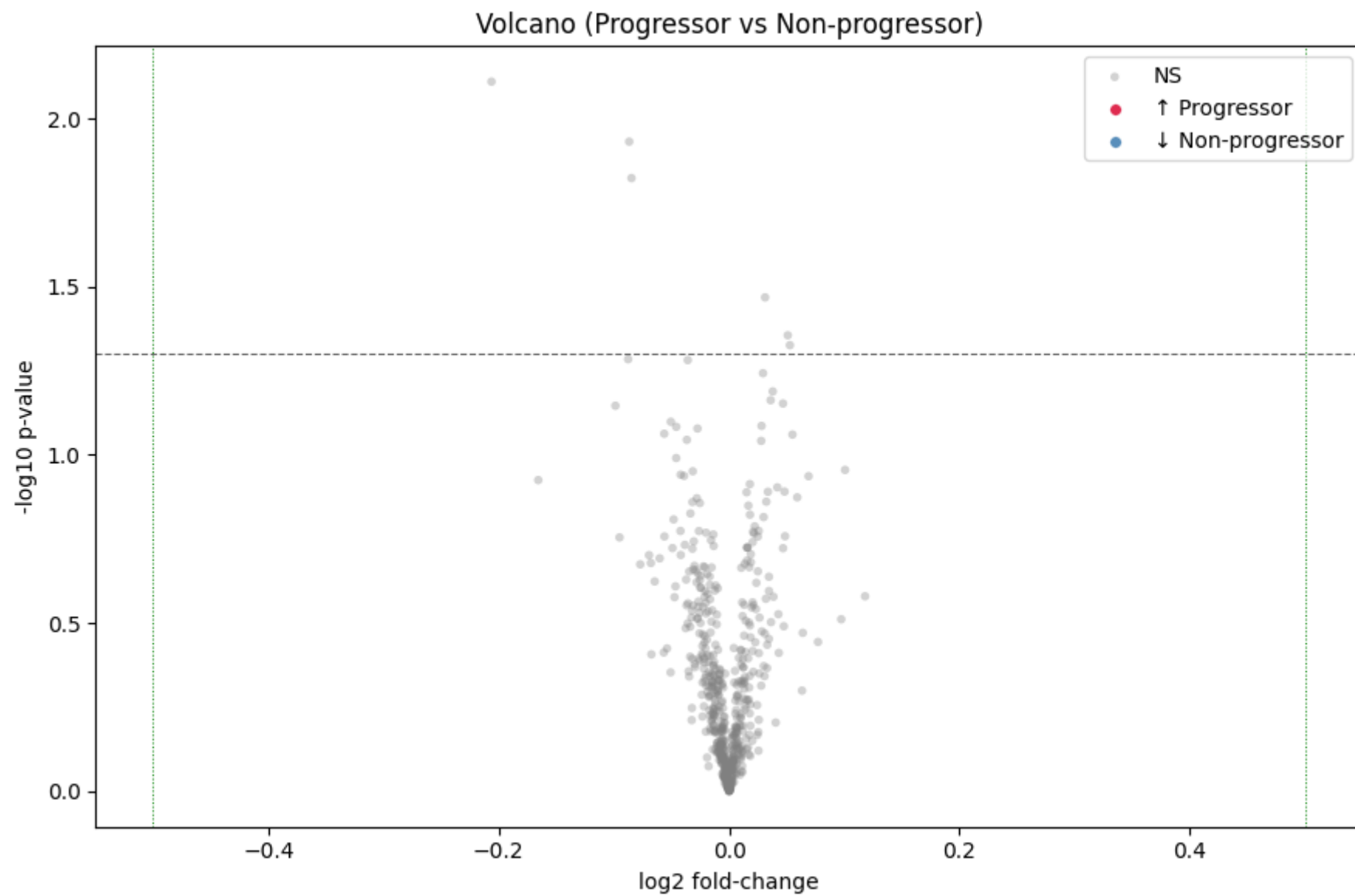
from src.metabolomics_univariate import differential_analysis
from src.metabolomics_plotting import plot_volcano, plot_volcano_categorized

diff = differential_analysis(
    Xb_log,
    metab["progressor_status"],
    group_a="Progressor",
    group_b="Non-progressor",
    include_cohen_d=True,
)
print(diff.head(15))
fig = plot_volcano(diff, alpha=0.05, fc_thresh=0.5)
fig.savefig("volcano_baseline_progressor.png", dpi=150, bbox_inches="tight")
figc = plot_volcano_categorized(
    diff,
    group_up="Progressor",
    group_down="Non-progressor",
    alpha=0.05,
    fc_thresh=0.5,
)
figc.savefig("volcano_categorized.png", dpi=150, bbox_inches="tight")
```


	feature	mean_Progressor	\
0	pos__plasmenyl-PE 36:2; [M+Na]+@7.41	5.911576	
1	pos__plasmenyl-PC 40:5; [M+Na]+@7.22	7.488497	
2	pos__TG 60:13; [M+NH4]+@9.81	7.408232	
3	pos__SM 42:5; [M]+@6.58	9.712194	
4	pos__DG 32:0; [M+NH4]+@7.98	9.318834	
5	pos__TG 42:2; [M+NH4]+@9.31	9.863338	
6	neg__PE 33:0; [M-H]-@7.08	6.951440	
7	neg__PC 35:4; [M-Ac-H]-@5.99	9.914988	
8	pos__lysoPC 16:1; [M+Na]+@1.17	9.342545	
9	pos__DG 34:2; [M+NH4]+@7.64	11.258055	
10	pos__TG 40:1; [M+NH4]+@9.23	10.420522	
11	pos__DG 35:2; [M+NH4]+@7.91	8.318187	
12	neg__PI 36:1; [M-H]-@6.37	8.260233	
13	neg__plasmenyl-PE 38:6; [M-H]-@6.64	11.449417	
14	pos__DG 36:3; [M+NH4]+@7.76	11.592364	

	mean_Non-progressor	log2fc	pvalue	cohens_d	padj
0	6.820332	-0.206299	0.007768	-0.611818	0.999384
1	7.952638	-0.086757	0.011713	-0.577652	0.999384
2	7.856581	-0.084772	0.015034	-0.556078	0.999384
3	9.504534	0.031181	0.034016	0.482577	0.999384
4	8.995738	0.050908	0.044087	0.457677	0.999384
5	9.509287	0.052739	0.047176	0.450940	0.999384
6	7.387032	-0.087683	0.051933	-0.442805	0.999384
7	10.165005	-0.035928	0.052248	-0.444052	0.999384
8	9.154197	0.029382	0.057159	0.432042	0.999384
9	10.966582	0.037844	0.064748	0.419211	0.999384
10	10.163453	0.036037	0.068761	0.412725	0.999384
11	8.053017	0.046740	0.070328	0.411581	0.999384
12	8.845312	-0.098731	0.071424	-0.412384	0.999384
13	11.858262	-0.050618	0.079686	-0.398481	0.999384
14	11.368539	0.028128	0.081975	0.394303	0.999384

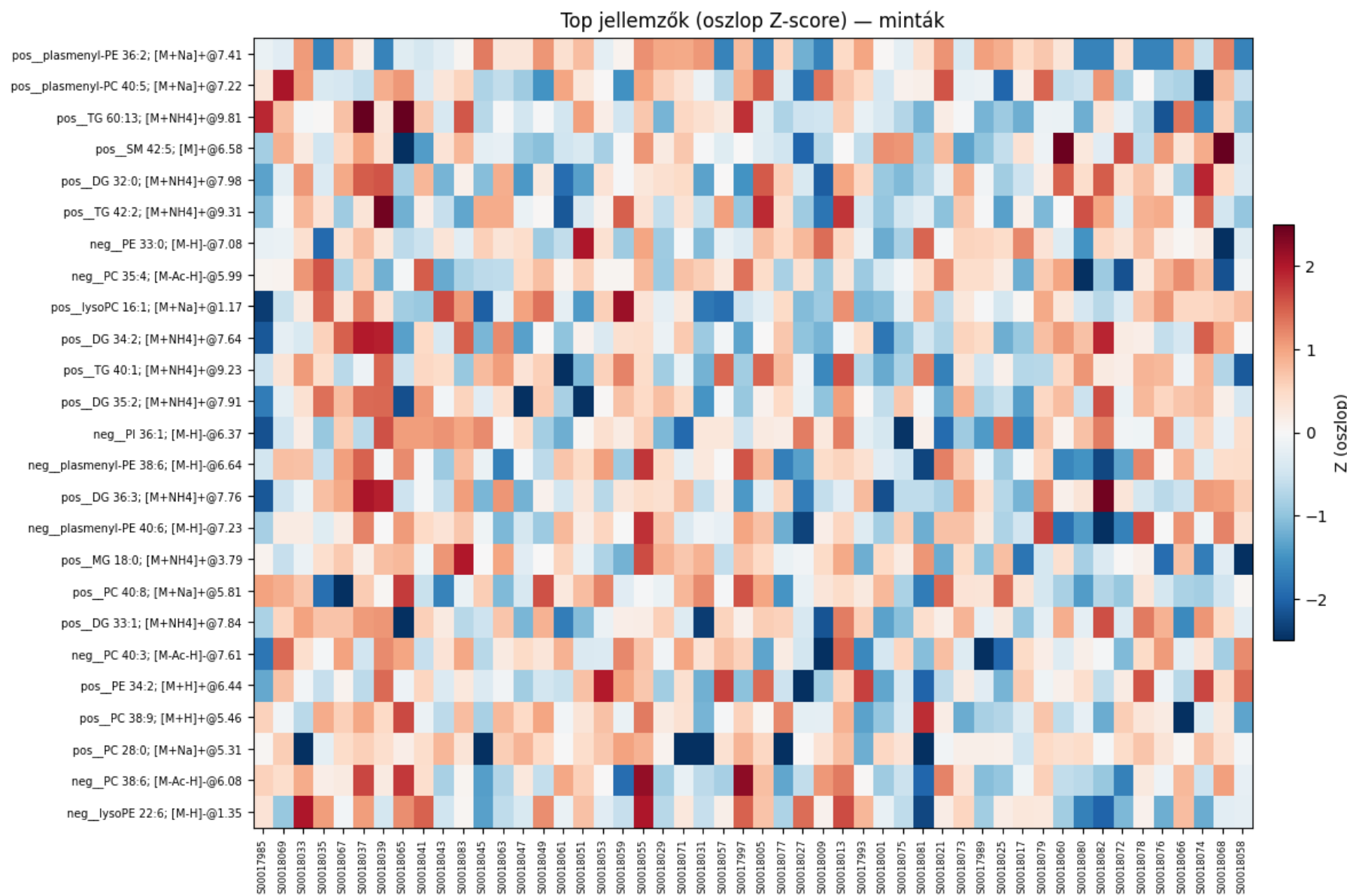




6. Top eltérések heatmap

```
In [31]: from src.metabolomics_plotting import plot_top_features_heatmap

top_n = 25
top_feats = diff.nsmallest(top_n, "padj")["feature"].tolist()
fig = plot_top_features_heatmap(
    Xb_par,
    top_feats,
    metab,
    group_col="progressor_status",
    max_samples=50,
)
fig.savefig("heatmap_top_features.png", dpi=150, bbox_inches="tight")
```



7. PLS-DA + VIP + S-plot (Pareto mátrix)

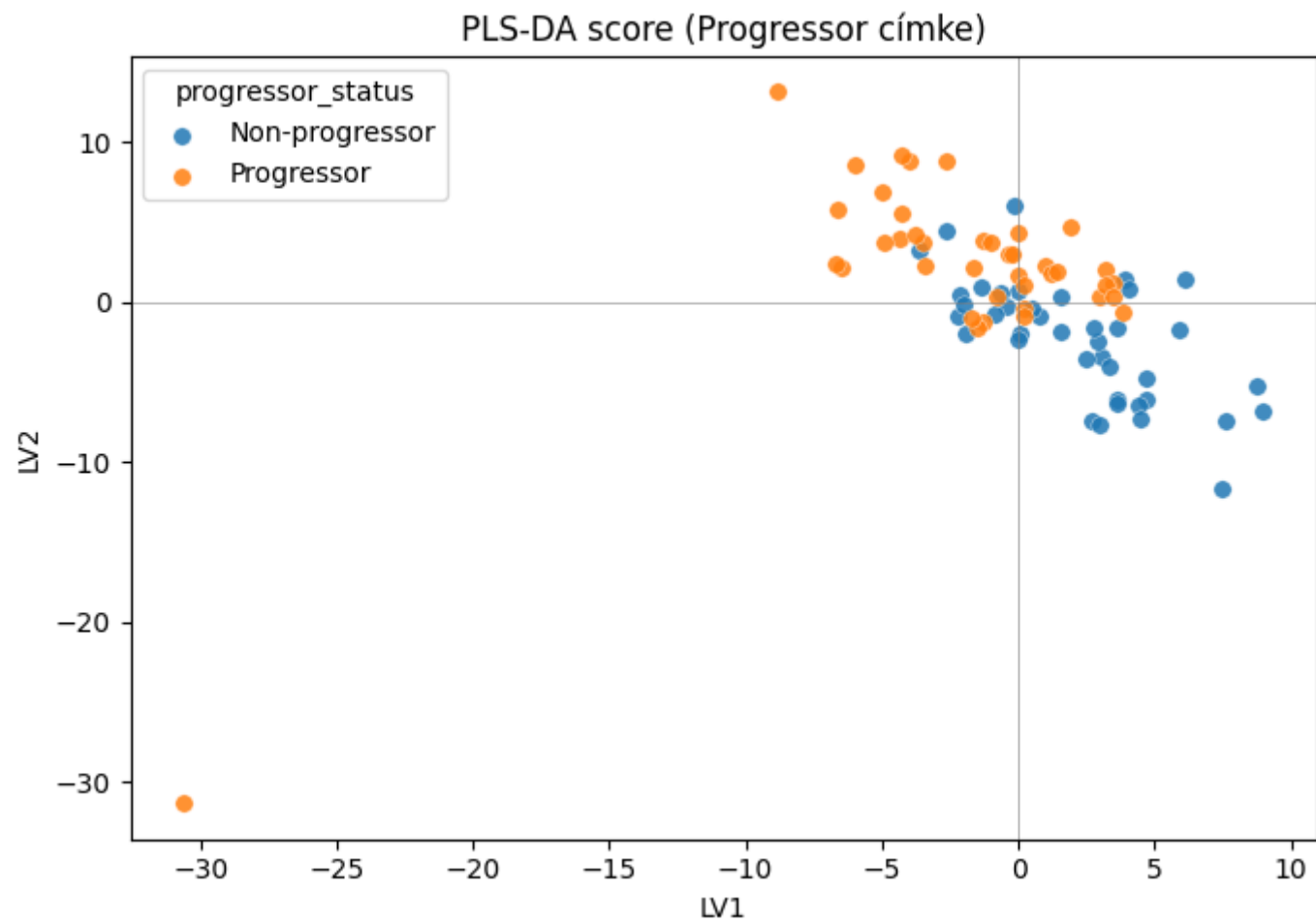
```
In [32]: from src.metabolomics_multivariate import fit_plsda_multiclass
from src.metabolomics_plotting import plot_s_plot_lv1
import matplotlib.pyplot as plt

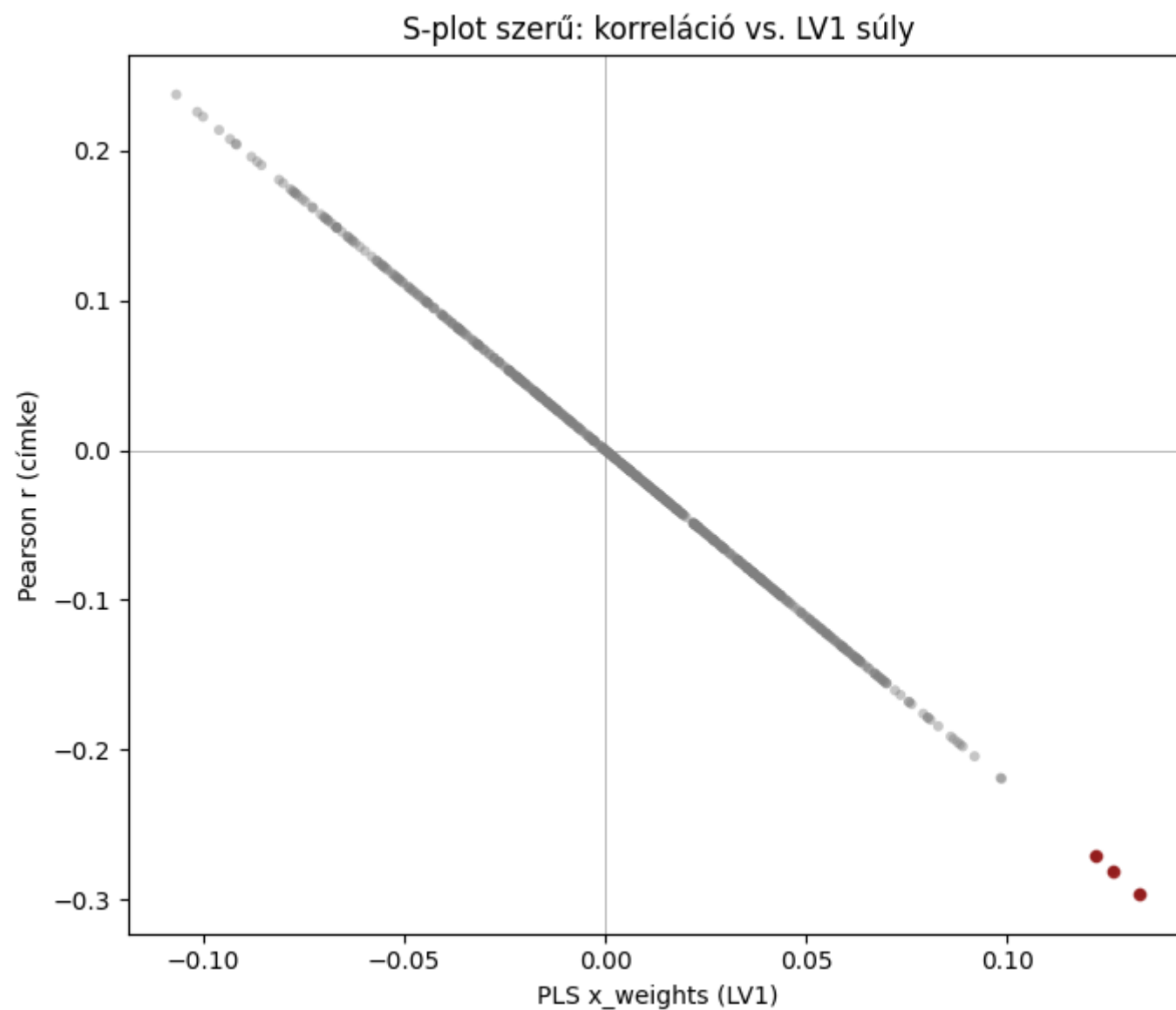
plsda = fit_plsda_multiclass(
    Xb_par,
    metab["progressor_status"],
    n_components=3,
    scale=True,
)
s_pls = plsda["scores"]
vip = plsda["vip"]
print(vip.nlargest(10))

fig, ax = plt.subplots(figsize=(7, 5))
for lab in metab["progressor_status"].unique():
    m = metab["progressor_status"] == lab
    ax.scatter(s_pls.loc[m, "LV1"], s_pls.loc[m, "LV2"], label=str(lab), s=45, alpha=0.85, edgecolors="white", line
ax.set_xlabel("LV1")
ax.set_ylabel("LV2")
ax.set_title("PLS-DA score (Progressor címke)")
ax.legend(title="progressor_status")
ax.axhline(0, color="gray", lw=0.4)
ax.axvline(0, color="gray", lw=0.4)
fig.tight_layout()
fig.savefig("plsda_scores.png", dpi=150, bbox_inches="tight")

w1 = plsda["model"].pls.x_weights[:, 0]
figs = plot_s_plot_lv1(Xb_par, metab["progressor_status"], "Progressor", w1)
figs.savefig("splot_lv1.png", dpi=150, bbox_inches="tight")

pos__DG 35:1; [M+NH4]+@8.27          2.536248
pos__TG 53:1; [M+NH4]+@10.97         2.177371
pos__plasmenyl-PE 36:2; [M+Na]+@7.41  1.983380
pos__TG 52:1; [M+NH4]+@10.76         1.964090
pos__plasmenyl-PC 40:5; [M+Na]+@7.22  1.933820
pos__TG 50:2; [M+NH4]+@10.39         1.919937
pos__TG 48:2; [M+NH4]+@10.17         1.895682
pos__TG 60:13; [M+NH4]+@9.81         1.889117
pos__DG 32:0; [M+NH4]+@7.98          1.884647
pos__IS DG 34:1; [M+NH4]+           1.852424
Name: VIP, dtype: float64
```





8. Random Forest — feature importance (felügyelt, gyors)


```
In [33]: from src.metabolomics_ml import random_forest_feature_importance
```

```
imp = random_forest_feature_importance(  
    Xb_par,  
    metab["progressor_status"],  
    "Progressor",  
    n_estimators=300,  
)  
print(imp.head(20))
```

```
neg__lysoPE 20:3; [M-H]-@1.68      0.013393  
pos__plasmenyl-PE 36:2; [M+Na]+@7.41 0.008696  
pos__PC 38:3; [M+H]+@7.11         0.008648  
pos__DG 38:2; [M+NH4]+@8.54       0.007971  
neg__PC 33:2; [M-Ac-H]-@6.03      0.007003  
pos__TG 42:2; [M+NH4]+@9.31       0.006971  
pos__PC 40:0; [M+H]+@7.31         0.006247  
pos__lysoPC 22:0; [M+H]+@4.31     0.005514  
pos__PE 38:4; [M+H]+@7.04         0.005253  
neg__plasmenyl-PE 38:6; [M-H]-@6.64 0.005251  
neg__lysoPE 22:1; [M-H]-@3.60     0.005209  
neg__plasmenyl-PE 34:3; [M-H]-@6.39 0.005051  
pos__MG 18:0; [M+NH4]+@3.79       0.004823  
pos__DG 32:0; [M+NH4]+@7.98       0.004475  
pos__TG 60:13; [M+NH4]+@9.81      0.004451  
neg__PC 42:4; [M-Ac-H]-@8.02      0.004386  
pos__IS D31-TAG                   0.004380  
neg__PC 40:3; [M-Ac-H]-@7.61      0.004191  
pos__PE 36:2; [M+H]+@7.03         0.004140  
pos__lysoPC 20:5; [M+H]+@1.04     0.004065  
Name: importance, dtype: float64
```

9. Rövid értelmezés

- **QC:** extrém `total_signal` vagy `missing_frac` minták kiszűrhetők.
- **PCA:** fő irányok a legnagyobb közös variancia mentén; színezés a klinikai címkével.
- **Vulkán + FDR:** egyszerre nézünk hatást ($\log_2\text{FC}$) és bizonytalanságot (p / FDR).
- **PLS-DA + VIP + S-plot:** felügyelt irányok és jellemző-súlyok (NMR notebook mintájára).
- **RF importance:** nemlineáris alternatíva a fontos peak-ekhez.

A `NMR_Metabolomika_Pipeline_v2.ipynb` referenciához igazítva: **PQN / log1p / Pareto, multipanel PCA, FDR (statsmodels BH), kategorizált vulkán, PLS-DA + VIP**; GPU/SHAP részek nélkül, LC–MS lipid adataira.